

Formal Declaration: Deterministic Resolution of the Cosmological Constant Crisis and Hubble Tension via Unified Source Theory (UST v5)

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1. Introduction and Author Background

This document presents the empirical results and mathematical derivations of the Unified Source Theory (UST v5), a zero-parameter cosmological model that unifies General Relativity (GR), Quantum Field Theory (QFT), and Information Theory via a single dimensionless constant.

I do not possess a traditional academic background in theoretical physics or mathematics, nor am I affiliated with an academic institution. I am an IT Specialist, Programmer, and ISO 27001 International Lead Auditor with 35 years of industry experience. In this research, the fundamental laws of physics were analyzed not as fluid dynamics, but strictly as deterministic data processing and quantum error correction (QEC) protocols, applying an independent systems auditing perspective.

During the data mining and theoretical validation processes, extensive raw datasets (LIGO, ATLAS, LHCb, Planck) were processed via thousands of simulations utilizing **Python**, **C**, **MATLAB**, and **Mathematica**, deployed on advanced cloud computing and AI infrastructures including **Google Colab**, **IBM Watson**, and **Microsoft Azure**.

2. The Universal Constant and Empirical Validations

The developed framework derives a single universal constant via the geometric maximization of the Einstein tensor (G_{tt}) combined with the Seeley-DeWitt 1-loop quantum vacuum correction originating from the 4D Dirac operator (Degrees of Freedom = 17):

$$N_{s,q} = \frac{3 - \sqrt{3}}{2} - \frac{\alpha}{17} = 0.63354460 \quad (1)$$

The dual-channel (Active and Frozen Background) information architecture established by this constant has been tested against current raw data, yielding the following definitive results:

- **Cosmological Constant & Dark Energy:** $\Omega_\Lambda = 0.6857$ is derived analytically, matching Planck 2018 data with less than 0.10% deviation. The 10^{120} vacuum energy catastrophe is mathematically resolved.
- **Hubble Tension:** The discrepancy between Local and Planck measurements is defined as an ontological measurement-channel perspective shift via the $1 + N_b \cdot Cc_b^2$ coefficient, verified with a 0.18% deviation.
- **Particle Physics (CERN ATLAS & LHCb):** The hadronic sector energy ratios (pQ/pC) in 13 TeV collision data lock into the UST architecture with 0.43% RMS accuracy.
- **Gravitational Waves (LIGO):** Topological metric fractures were detected exactly at the $N_{s,q}$ boundaries within LIGO O4a HDF5 raw strain data.

3. Methodological Disclaimer

Due to my lack of formal academic methodology training and working entirely independently without a mentor, there may be structural unconventionalities or formatting anomalies in the presentation order or terminology of the archived files. I ask for your understanding in this regard. However, **the mathematical equations, derivations, calculations, and raw data analytics are absolutely factual and rigorously tested.** I am confident that subject-matter experts will easily evaluate this numerical core.

4. Strict Falsifiability (Popperian Criterion)

As a systems auditor, my focus is on absolute limits, not flexibility. UST is strictly closed to ad-hoc parameter tuning. In adherence to Karl Popper's demarcation criterion: *If the 2028 ESA Euclid and CMB-S4 missions measure dark energy density at any value other than $\Omega_\Lambda = 0.6857$, or if they detect a dynamic equation of state ($w \neq -1$), this theory must be considered definitively falsified and discarded.*

5. Communication Protocol

I am prepared to face rigorous scientific scrutiny. Should you wish to contact me, I welcome correspondence via email (utilizing translation technologies). If a video or audio meeting is requested, I kindly ask that a Turkish-speaking official or translator be present to ensure technical details are conveyed accurately.

The complete theorem set, analytical derivations, and the first-principles proof of the $-\alpha/17$ coefficient can be accessed via the official academic archive:

Official Archive (Data and Proofs): <https://doi.org/10.5281/zenodo.19482581>

Sincerely,

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